Determine the Area of Trapezoids

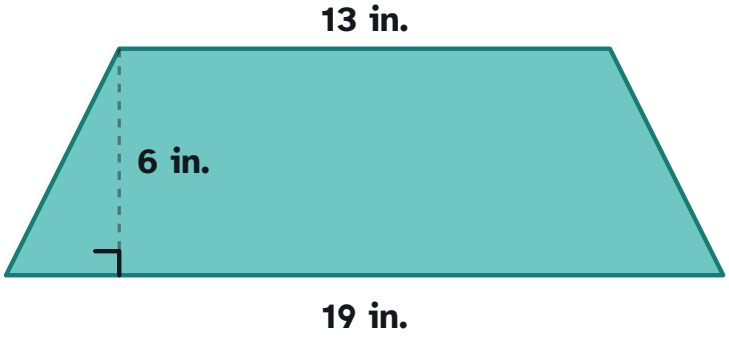
Use the trapezoid formula, and know why it has a half.

Name _____ **Date** _____

Tonight's spark

A sewing club cuts the side of a tote bag. It is a trapezoid. The top is 13 inches, the bottom is 19 inches, and it is 6 inches tall. How much cloth is that?

Guess first: is the cloth more or less than 100 square inches?



Our class model: two trapezoids make a parallelogram

Try it

Work it out on paper. Write one answer on each line.

1 **In A = ½ × (b1 + b2) × h, what do you do with the two bases first?**

A. Subtract b2 from b1 **B.** Multiply b1 by b2 **C.** Multiply them by h **D.** Add them

My answer _____

2 **Why is there a ½ in the trapezoid area formula?**

A. Two identical trapezoids make one parallelogram, so one trapezoid is half

B. Every trapezoid has two bases **C.** A trapezoid has half as many sides as a rectangle

D. The height is always cut in half

My answer _____

3 **A trapezoid has bases of 3 and 9 and a height of 4. What is its area?**

A. 48 **B.** 54 **C.** 24 **D.** 16

My answer _____

4 **A trapezoid has bases of 7 and 5 and a height of 6. What is its area?**

A. 72 **B.** 36 **C.** 18 **D.** 210

My answer _____

5 **A trapezoid has bases of 10 and 6 and a height of 4. The product (10 + 6) × 4 is 64. What is the area of the trapezoid?**

A. 64 square units **B.** 16 square units **C.** 24 square units **D.** 32 square units

My answer _____

Talk about it

Which two sides are the bases, and why do we add them before multiplying by the height?

Home mission

Cut two same trapezoids from scrap paper. Flip one over and fit the two together. What shape do they make? Can your student say why that explains the ½?

If we get stuck, ask

- Which two sides are parallel? Those are the bases.
- What is inside the parentheses? Do that first.
- Did you take half of your last answer?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-3-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it